

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: Recombinant pMAM neo vectors (8.8 to 9.2 kB) - confirm via expression (Northern blot) immunoblotting, PCR & Southern blot
Species Used	Human, squamous cell carcinoma lines, oral & cervical; Monkey, Vero, kidney cells.		

Before the Pulse

Cell Growth Medium	M199, MEM, DMEM/F12 (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Mid-log phase
		Pre-pulse Incubation	Ice, 10 min., in media with DNA.
Wash Solution	Ice-cold Phosphate Buffered Saline, pH 7.2		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	Growth media	Voltage	0.250 kV
Cell Density	1 x 10 ⁷ (7) cells / ml	Field Strength	0.625 kV/cm
Volume of Cells	3 x 10 ⁶ (6) cells / pulse	Capacitor	500 µF
DNA Concentration	Not given	Resistor	(Pulse Controller) none Ω
DNA Resuspension Buffer	Not given	Time Constant	8 to 12 msec
Volume of DNA	100 µl		
After the Pulse			
Outgrowth Medium	Growth media, plus or minus G418		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37°C
Length of Incubation	at least 10 days
Selection Method or Assay Used	G418
Electroporation Efficiency	6 x 10 ⁻⁵ to 9 x 10 ⁻⁸ ; depends on cell line used
Per Cent Survival	50 %

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Survey Number

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